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ANATOMY

Pedicle Morphometry of Lumbar Vertebrae

Male, Taller, and Heavier Specimens Have Bigger Pedicles

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Study Design. An anatomic study of pedicle dimensions was performed for lumbar vertebrae from American subjects.

Objective. To quantify the dimensions of the lumbar pedicles and to better define the demographic factors that could ultimately govern the caliber selection of pedicle screws.

Summary of Background Data. Transpedicular screw fixation allows for segmental instrumentation into multiple vertebrae across multilevel fusion area, offering considerable biomechanical advantage over the conventional hook and lateral mass fixation. Large variations in morphology from previous studies may be related to differences in demographics, sample size, and methodology.

Methods. For this study, L1–L5 vertebrae from 503 American human cadavers were directly measured with a digital caliper. Examiner measured each vertebra to determine medial–lateral pedicle width (PW) and cranial–caudal pedicle height (PH). Demographic information regarding age, sex, and race, as well as body height and weight, was available for all 503 subjects.

Results. PH decreased in size caudally down the lumbar spine, but PW increased in size. The largest PH was at the L1 level with a mean of 15.75 mm. The widest PW was at the L5 level with a mean of 18.33 mm. Males have larger pedicles than females for all lumbar levels. The tallest and heaviest groups generally had larger pedicles than the shorter and lighter groups, respectively. Age and race did not consistently affect pedicle dimension in a statistically significant manner.

Conclusion. Our large-scale study of American specimens characterized the relationship between pedicle dimensions and a variety of demographic factors such as age, sex, body height, and weight. With substantial statistical power, the current study showed that male, taller, and heavier individuals had larger lumbar pedicles.

Key words: age, body heightv body weight, demographic factors, lumbar, morphometry, pedicle height, pedicle width, scoliosis, transpedicular fixation.

Level of Evidence: 3

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Posterior spinal instrumentation using hooks, rods, and wires played an important role in the revolution of spinal fixation. Harrington and Luque systems spanned multiple vertebral levels to establish the 3-dimensional link between implant and vertebral body.[1,2] This method leads to significant reduction of spine mobility, which is particularly relevant in the lumbar region.[2] Transpedicular fixation allows for segmental instrumentation into multiple vertebrae across multilevel fusion area, thus offering considerable biomechanical advantage over the conventional hook fixation and lateral mass screw fixation.[3–6]

The pedicle has been proven to be the strongest part of the vertebra even in osteoporotic bone.[7–9] Several systems of transpedicular fixation are available today that take advantage of this special quality of pedicles.[7,10–12] Despite its true 3-column fixation and superior pullout strength, screw fixation in the lumbar region has been controversial due to reported size variation in the lumbar pedicles. Ofiram *et al* argued that it is safer to place screws in the lower thoracic spine than in the upper lumbar spine.[13]

Numerous similar morphometric studies have surfaced in the literature during the past 2 decades. Researchers used imaging modalities including CT and MRI,[13–16] as well as direct measurements using digital calipers[17–19] to assess pedicle anatomy. Olsewski *et al* compared direct and radiographical measurements and found significant differences.[20] The lumbar spine has a natural lordosis that must be preserved for function. This, in addition to the pedicle's complicated inner structure and special spatial orientation, prevents accurate preoperative planning using imaging modalities. Therefore, direct caliper measurements provide

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the most accurate account of pedicle morphometry. Additionally, most studies have very small sample size, ranging from 10 specimens[21] up to 180 specimens.[22] Differences in methodology and sample size significantly contribute to published variations in pedicle morphometry. Our study directly measures 503 cadaveric specimens from an American population to provide the most accurate and precise account of lumbar pedicle morphometry. Although the differences in pedicle size between males and females have been well documented,[13,16,18,20] our study further explores the effects of age, race, and height on pedicle morphometry. We previously described both upper thoracic (T1–T6) and lower thoracic (T7–T12) pedicle morphometry.[23,24] These findings about the relationship of pedicle dimension to different demographic factors can assist the spinal surgeon in adequate preoperative planning.

MATERIALS AND METHODS

Subjects

The Hamann-Todd Osteological Collection in Cleveland, Ohio, contains more than 3300 treated and dried specimens, making it the largest modern collection of documented human skeletons in the world. The specimens in the collection are from individuals who died in Cleveland between 1893 and 1938. Specimens from this collection include documented height and weight at the time of death, sex, and ethnic origin. For this study, 503 specimens were randomly chosen for examination in no particular order. Those with congenital diseases such as scoliosis were excluded; documentation of past medical and surgical history was largely unavailable. We included 404 male and 99 female specimens; 222 were from African American individuals and the remainder from Caucasian individuals. The average age of the sample cohort was 47 years (in the range from 17 to 93 yr). The average height was 170 cm and the average weight was 56 kg. Demographic data pertaining to age, sex, and ethnicity of the 503 specimens were documented (Table 1).

1 examiner (C.C.Y.) subjectively measured the gross specimens by digital caliper with a precision of 100th of a millimeter. The caliper was placed at the narrowest points of the pedicle height (PH) and pedicle width (PW) along the longitudinal axis. PH was measured with a lateral approach of the caliper (Figure 1A) in the sagittal plane, and PW was measured with an inferior approach (Figure 1B) in the transverse plane. Measurements on all vertebrae of 1 specimen were completed before moving onto the next specimen.

Statistical Analyses

Means and standard deviations were determined for each of the measurements made for the population as a whole and for various subpopulations, based on age, sex, race, height, and weight. A multiple regression model was developed to compare pedicle dimension differences with all specimen demographic data (age, sex, race, height, and weight). Statistical analyses were performed using Statgraphics Centurion XVI (version 16.2; Statpoint Technologies Inc., Warrenton, VA). One-way analysis of variance (ANOVA) and Student *t* tests were used to determine statistical significance. Differences in means with $P < 0.05$ were considered statistically significant.

RESULTS

The averages of left and right pedicles were calculated and used for data analysis. Left and right measurements were identical ($P > 0.05$) except at the following levels ($P < 0.05$): L1 PH (left, 15.57 mm *vs.* right, 15.94 mm) and L2 PH (left, 14.96 mm *vs.* right, 15.17 mm). On the basis of 21 random samples, the averaged inter- and intrarater coefficients of variation were 5.2% (1.5%–16.8%) and 4.0% (1.7%–6.2%), respectively. PH generally decreased in size caudally down the lumbar spine (Figure 2). The highest PH was at the L1 level with a mean of 15.75 mm (range, 10.84–22.33 mm), whereas the lowest one was at the L5 level with a mean of 13.67 mm (range, 9.72–19.94 mm). PW generally increased in size caudally down the lumbar spine (Figure 2). The widest PW was at the L5 level with a mean of 18.34 (range, 9.09–29.06 mm), whereas the narrowest one was at the L1 level with a mean of 8.53 mm (range, 3.92–18.74 mm). The percentage of pedicles with PW less than 4.5 mm included: L1, 0.80% and L2, 0.40%. Similarly, the

TABLE 1. Age, Sex, and Racial Breakdown of all the Sampled Specimens

Age (yr)	Number of Specimens	Female	Male	Caucasian	African American
15–24	35	12	23	3	32
25–34	73	21	52	24	49
35–44	114	25	89	57	57
45–54	129	18	111	78	51
55–64	77	7	70	59	18
65–74	45	8	37	35	10
75–84	24	4	20	22	2
85–94	6	4	2	3	3
Total	503	99	404	281	222

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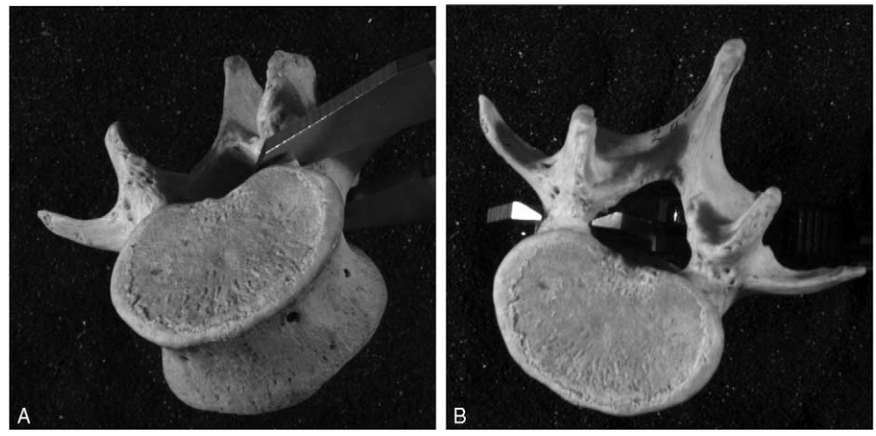


Figure 1. Lumbar pedicle height (PH) (A) is measured with a lateral approach. Pedicle width (PW) (B) is measured with an inferior approach. The average of both pedicles was used in the study.

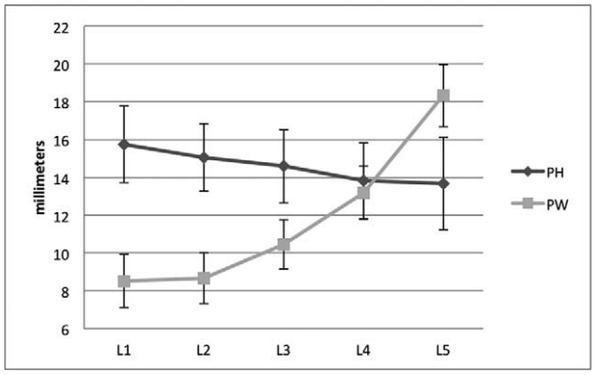


Figure 2. Lumbar PH and PW. Values are mean ± SD. PW indicates pedicle width; PH, pedicle height.

percentage less than 5.5 mm included: L1, 4.77%; L2, 2.78%; and L3, 0.20%.

Age

Linear regressions of PH and PW *versus* age calculated the coefficients at 0.008 to 0.026 mm/yr (Table 2, column A) after correcting for sex, race, height, and weight. Comparing a 25-year-old person with a 75-year-old person, a lengthening of 0.40 to 1.3 mm could be expected in PH, and a widening of 0.55 to 0.90 mm in PW. With respect to age, specimens were categorized into 4 quartiles: 30 years or less, 31 to 49 years, 50 to 69 years, and 70 years or more (Table 3). Generally, pedicle dimension increased proportionally with age until the eldest quartile range, when dimensions decreased, except for L5 in which the eldest quartile range continued to increase in pedicle dimensions. ANOVA analysis showed significant *P* values for all lumbar levels except PH of L3, and PW of L1, L2, and L3. In addition, the number of subjects with PW less than 4.5 mm was plotted on the basis of the 4 quartiles of age (Figure 3A). In levels L3 through L5, virtually no PW was less than 4.5-mm wide, regardless of subage groups. For the L1 level, the eldest age group had the largest percentage of pedicles with PW less than 4.5 mm (1.92%), the 50- to 69-year age group had a lesser percentage of pedicles with PW less than 4.5 mm (1.30%), the 31- to 49-year age group had an even smaller percentage of pedicles with PW less than 4.5 mm (0.47%), and the youngest quartile having virtually no pedicles with PW less than 4.5 mm. For the L2 level, the 50- to 69-year age group had the largest percentage of pedicles with PW less than 4.5 mm (0.65%), the 31- to 49-year age group had a lesser percentage of pedicles with PW less than 4.5 mm (0.47%), and the youngest and eldest quartile age groups had virtually no pedicles with PW less than 4.5 mm.

TABLE 2. A to C, Linear Regression Coefficients of Pedicle Dimensions *vs.* Age (in yr, A), *vs.* Body Height (in m, B), and *vs.* Body Weight (in kg, C)

A	PH (mm/yr)	PW (mm/yr)	B	PH (mm/m)	PW (mm/m)	C	PH (mm/kg)	PW (mm/kg)
L1	0.017	0.008	L1	5.51	2.07	L1	0.001	0.017
L2	0.011	0.009	L2	4.97	2.71	L2	0.005	0.022
L3	0.012	0.012	L3	5.56	3.34	L3	−0.004	0.015
L4	0.016	0.015	L4	5.02	4.20	L4	−0.003	0.015
L5	0.018	0.026	L5	5.68	7.59	L5	−0.011	0.015

Y-intercepts for L1–L5 PH are 4.95, 5.26, 4.43, 4.26, and 3.33 mm, respectively.
Y-intercepts for L1–L5 PW are 2.84, 1.59, 2.35, 3.44, and 2.17 mm, respectively.
Data obtained after correcting for sex, race, body height, and weight (A), for age, sex, race, and body weight (B), and for age, sex, race, and body height (C).
PW indicates pedicle width; PH, pedicle height.

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TABLE 3. PH and PW (Mean ± SD) of All the Sampled Specimens, Categorized into 4 Quartiles Based on Age, Height, and Weight

Age (yr)	No. of Specimens	L1 PH (mm)	L2	L3	L4	L5
≤30	82	15.35 ± 1.26	14.71 ± 1.31	14.30 ± 1.33	13.55 ± 1.41	13.49 ± 1.53
31–49	215	15.68 ± 1.39	15.02 ± 1.33	14.55 ± 1.32	13.68 ± 1.39	13.49 ± 1.55
50–69	154	16.05 ± 1.47	15.35 ± 1.31	14.78 ± 1.34	14.08 ± 1.40	13.92 ± 1.83
≥70	52	15.84 ± 1.39	14.96 ± 1.24	14.62 ± 1.15	14.00 ± 1.45	14.00 ± 1.67
P		2.37E−03	2.96E−03	0.06	0.01	0.03
Age (yr)	**No. of Specimens**	**L1 PW (yr)**	**L2**	**L3**	**L4**	**L5**
≤30	82	8.41 ± 1.86	8.46 ± 1.73	10.12 ± 1.80	12.62 ± 1.95	17.32 ± 2.16
31–49	215	8.48 ± 1.98	8.63 ± 1.65	10.37 ± 1.73	13.14 ± 1.78	18.25 ± 2.25
50–69	154	8.68 ± 2.10	8.83 ± 1.99	10.73 ± 2.20	13.50 ± 2.19	18.74 ± 2.55
≥70	52	8.45 ± 2.26	8.67 ± 1.82	10.60 ± 2.07	13.37 ± 2.21	19.15 ± 2.54
P		0.73	0.49	0.10	0.01	1.17E-05
Height (mm)	**No. of Specimens**	**L1 PH (mm)**	**L2**	**L3**	**L4**	**L5**
≤1600	69	14.55 ± 1.03	14.03 ± 1.16	13.55 ± 1.08	12.75 ± 1.25	12.65 ± 1.47
1601–1699	168	15.42 ± 1.38	14.77 ± 1.24	14.35 ± 1.23	13.60 ± 1.43	13.55 ± 1.75
1700–1799	204	16.17 ± 1.26	15.42 ± 1.18	14.89 ± 1.21	14.13 ± 1.24	13.82 ± 1.44
≥1800	62	16.62 ± 1.18	15.85 ± 1.27	15.40 ± 1.26	14.53 ± 1.28	14.68 ± 1.61
P		<1E−10	<1E−10	<1E−10	4.44E−16	1.11E−11
Height (mm)	**No. of Specimens**	**L1 PW (mm)**	**L2**	**L3**	**L4**	**L5**
≤1600	69	7.55 ± 2.01	7.47 ± 1.89	9.26 ± 2.04	11.74 ± 1.94	16.65 ± 2.33
1601–1699	168	8.27 ± 2.03	8.48 ± 1.77	10.29 ± 1.87	13.03 ± 2.02	18.08 ± 2.41
1700–1799	204	8.84 ± 1.83	8.98 ± 1.57	10.81 ± 1.81	13.61 ± 1.80	18.83 ± 2.19
≥1800	62	9.29 ± 2.16	9.44 ± 1.74	11.12 ± 1.81	13.86 ± 1.81	19.32 ± 2.20
P		2.56E−07	1.84E−11	2.24E−09	3.89E−12	2.23E−12
Weight (kg)	**No. of Specimens**	**L1 PH (kg)**	**L2**	**L3**	**L4**	**L5**
≤45	83	15.32 ± 1.43	14.46 ± 1.29	14.15 ± 1.32	13.47 ± 1.48	13.43 ± 1.63
46–54	123	15.58 ± 1.38	14.95 ± 1.29	14.56 ± 1.25	13.70 ± 1.39	13.74 ± 1.72
55–63	206	15.88 ± 1.39	15.23 ± 1.30	14.73 ± 1.31	13.96 ± 1.35	13.68 ± 1.62
≥64	91	16.09 ± 1.39	15.40 ± 1.31	14.72 ± 1.34	13.94 ± 1.48	13.80 ± 1.68
P		0.00081	4.15E−06	0.0052	0.030	0.47
Weight (kg)	**No. of Specimens**	**L1 PW (kg)**	**L2**	**L3**	**L4**	**L5**
≤45	83	7.94 ± 2.00	7.86 ± 1.67	9.78 ± 1.90	12.46 ± 2.06	17.39 ± 2.33
46–54	123	8.17 ± 1.81	8.49 ± 1.60	10.28 ± 1.67	12.93 ± 1.87	18.02 ± 2.10
55–63	206	8.76 ± 1.95	8.89 ± 1.78	10.66 ± 2.00	13.41 ± 1.99	18.68 ± 2.48
≥64	91	9.01 ± 2.29	9.13 ± 1.92	10.87 ± 2.01	13.68 ± 1.95	18.87 ± 2.47
P		0.00029	3.68E−06	0.00047	0.00009	0.00003

PW indicates pedicle width; PH, pedicle height.

Body Height

Linear regressions of PH and PW *versus* body height calculated the coefficients at 2.07 to 7.59 mm/m (Table 2, column B) after correcting for age, sex, race, and weight. Comparing a 1.5-m tall person with a 2.0-m tall person, a lengthening of 2.49 to 2.84 mm could be expected in PH, and a widening of 1.04 to 3.80 mm could be expected in PW. With respect to body height (mm), specimens were categorized into 4 quartiles: 1600 mm or less, 1601 to 1699 mm, 1700 to 1799 mm, and 1800 mm or more (Table 3). Again, pedicle dimension increased proportionally with height. PH was clearly highest in the tallest group and lowest in the shortest group. PW was also widest in the tallest group and narrowest in the shortest group. ANOVA analysis showed significant *P* values for all lumbar levels. In addition, the number of subjects with PW less than 4.5 mm was plotted for each of the 4 quartiles of height (Figure 3B). In levels L3 through L5, virtually no PW was less than 4.5-mm wide, regardless of subheight groups.

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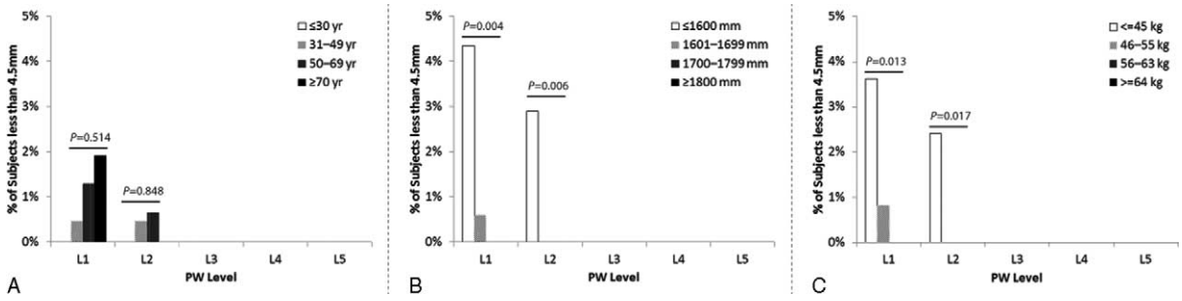


Figure 3. (A–C) Percentage of pedicles with PW, less than 4.5 mm, categorized into 4 quartiles based on age (in yr, A), body height (in mm, B), and body weight (in kg, C). χ^2 analysis was performed for each category (age, height, and weight) from L1 to L2; *P* values are shown.

For levels L1 and L2, the shortest group had the largest percentages of pedicles with PW less than 4.5 mm (4.35% and 2.90%, respectively), and the 1601- to 1699-mm group had the next smallest percentage (0.60% and 0%, respectively).

Body Weight

Linear regressions of PH and PW *versus* body weight were also calculated (Table 2, column C) after correcting for age, sex, race, and height. Comparing a 55-kg (121 lb) person with a 65-kg (143 lb) person, a shortening of 0.11 to a lengthening of 0.05 mm could be expected in PH, and a widening of 0.15 to 0.22 mm could be expected in PW. With respect to body weight (kg), specimens were categorized into 4 quartiles: 45 kg or less, 46 to 54 kg, 55 to 63 kg, and 64 kg or more (Table 3). PH increased proportionally with weight in the L1 and L2 levels. In the L3 and L4 levels, PH increased proportionally with weight until the last quartile, when it decreased. In the L5 level, PH increased proportionally with weight except for the 55- to 63-kg quartile, when it decreased. PW increased proportionally with weight, with PW highest in the heaviest group and lowest in the lightest group. ANOVA analysis showed significant *P* values for all lumbar levels, except for the L5 PH level. In addition, the number of subjects with PW less than 4.5 mm was plotted for each of the 4 quartiles of weight (Figure 3C). In levels L3 through L5, virtually no PW was less than 4.5-mm wide, regardless of sub-weight groups. For levels L1 and L2, the lightest group had the largest percentages of pedicles with PW less than 4.5 mm (3.61% and 2.41%, respectively), and the 46- to 55-kg group had the next smallest percentage (0.81% and 0%, respectively).

Sex and Race

With respect to sex, males had larger PH and PW than females from levels L1 to L5 (Table 4). Student *t* test showed

TABLE 4. PH and PW (Mean ± SD) of All the Sampled Specimens, Categorized by Sex and Race

Sex	No. of Specimens	L1 PH (mm)	L2	L3	L4	L5
Male	404	16.04 ± 1.35	15.33 ± 1.27	14.80 ± 1.28	14.05 ± 1.34	13.90 ± 1.65
Female	99	14.57 ± 0.95	13.98 ± 0.99	13.73 ± 1.07	12.85 ± 1.27	12.77 ± 1.36
P		1.25E–22	3.05E–21	6.28E–14	5.49E–15	4.59E–10
Sex	**No. of Specimens**	**L1 PW (mm)**	**L2**	**L3**	**L4**	**L5**
Male	404	8.88 ± 1.96	9.01 ± 1.69	10.84 ± 1.85	13.58 ± 1.89	18.78 ± 2.33
Female	99	7.06 ± 1.61	7.24 ± 1.47	8.91 ± 1.45	11.60 ± 1.59	16.47 ± 1.95
P		8.54E–17	1.92E–20	1.41E–20	2.61E–20	1.51E–18
Race	**No. of Specimens**	**L1 PH (mm)**	**L2**	**L3**	**L4**	**L5**
Caucasian	281	15.76 ± 1.40	15.11 ± 1.26	14.59 ± 1.25	13.80 ± 1.44	13.75 ± 1.62
African American	222	15.74 ± 1.42	15.01 ± 1.42	14.58 ± 1.39	13.82 ± 1.37	13.58 ± 1.69
P		0.87	0.42	0.92	0.89	0.25
Race	**No. of Specimens**	**L1 PW (mm)**	**L2**	**L3**	**L4**	**L5**
Caucasian	281	8.24 ± 1.90	8.50 ± 1.74	10.43 ± 1.93	13.22 ± 1.98	18.69 ± 2.40
African American	222	8.88 ± 2.12	8.86 ± 1.84	10.50 ± 1.95	13.14 ± 2.03	17.86 ± 2.41
P		3.99E-04	0.027	0.70	0.65	1.24E-04

PW indicates pedicle width; PH, pedicle height.

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significant *P* values for all lower lumbar levels. In all lumbar levels except L4, Caucasians had higher PH than African Americans. In L4, African Americans had larger PH than Caucasian. However, none of the levels reached statistical significance with regards to PH. In levels L1 through L3, African Americans had larger PW than Caucasians. In levels L4 and L5, Caucasians had larger PW than African Americans. All lumbar levels regarding PW reached statistical significance except for L3 and L4.

DISCUSSION

Pedicle screw placement in the lumbar region has become ubiquitous.[7,25] The increased application of transpedicular screw instrumentation warranted studies that can better define pedicle dimensions for spinal surgeons. Although many lumbar mormphometric measurements have been done, none of them has the statistical power or the population relevance like our study. The primary goal of the current study was to quantify the dimensions of the lumbar pedicles and to better define the demographic factors that could ultimately govern the caliber selection of pedicle screws.

Lumbar pedicle morphometry has been studied extensively during the past 2 decades, either directly or by radiological measurement.[13–21] The differing sample sizes, methods, and populations of the studies contributed to the wide disparity in the reported results. The best method of determining true pedicle dimensions is by direct and careful measurement of cadaveric specimens. Transverse and sagittal pedicle angles (TPA and SPA) are also important in operative planning and execution. However, we did not study them as they have been measured in great detail in the literature with goniometer[17,21,26] and imaging.[2] In the current study, all levels of the lumbar spine of 503 specimens were analyzed using direct caliper measurement. PW is the limiting factor in determining screw caliber selection due to its smaller size compared with PH. PW exhibited a widening pattern from L1 to L5. A comparison of the current study results with those of other morphometrical studies is shown (Table 5).

Previous studies of the relationships among demographic factors (age, sex, and race) and pedicle dimensions have been inconclusive. For example, Kim *et al*[19] demonstrated smaller pedicles in Koreans than in Westerners, and Chadha *et al*[14] and Acharya *et al*[15] showed significant pedicle size differences between Indian and Caucasian populations. Similar to Hou *et al*[18] and Olsewski *et al*[20] studies, the current study showed that males had significantly larger pedicle dimensions than females. From L1 to L5, PH between Caucasians and African Americans is not statistically different. However, compared with that of African Americans, PW of Caucasians is bigger at L1 and L2 but smaller at L5. With respect to age, the current study showed that PH generally increased with age, but PW failed to consistently demonstrate statistically significant differences among the 4 subage groups. Therefore, we cannot reliably use age cutoffs to predict the caliber selection of pedicle screws.

This study is the first of its kind to analyze how pedicle dimension varies with body weight. Both PH and PW increased with weight. Compared with that of the lightest group, PH of the heaviest group was as much as 0.94 mm higher, and PW was as much as 1.48 mm wider. The increased weight bearing on the skeletal system by heavier individuals could contribute to stronger bone quality, thus leading to better bone purchase by the pedicle screws. To validate the similarity in pedicle dimensions between contemporary population and people who died 100 years ago, morphometric data from 3 recent American studies[2,17,20] were analyzed. In general, the mean values of pedicle dimension are similar between the current study and these 3 studies (Table 5), which suggests that pedicle dimensions of the current study have not changed significantly when compared with those of the contemporary population.

Furthermore, body height is an important but often overlooked predictor of pedicle dimensions. Again, this is the first study of its kind to explore the effect of this variable on lumbar pedicle morphometry. The results from the current study showed that both PH and PW increased with height. Compared with that of the shortest group, PH of the tallest group was as much as 2.07 mm higher, and PW was as much as 2.67 mm wider.

Most spine surgeons prefer to use the largest fully threaded screw that can be safely accepted by a pedicle. A discrepancy between pedicle dimensions and screw caliber may lead to disruption of the pedicle's medial and lateral cortex, which can cause spinal canal penetration and nerve root injury, respectively. Understanding the factors that define the morphology of the lumbar pedicles and choosing the correct pedicle screw, therefore, would be the key for successful transpedicular screw instrumentation. The current study quantitatively stratifies the effects of demographic factors and body height and weight on pedicle dimensions. Virtually all lumbar levels can accept the smallest screw, at 4.5 mm. Figure 3 showed that the likelihood of accepting a 4.5-mm pedicle screw is significantly elevated with increasing height and weight; a similar trend was seen for the 5.5-mm screw. With our data, we can extrapolate sensitive cutoffs of demographic characteristics that would predict a more successful outcome for a patient undergoing a lumbar transpedicular procedure.

CONCLUSION

This large study of lumbar vertebrae of 503 American specimens characterized the relationship between pedicle dimensions and a variety of demographic factors. Sex, body height, and weight consistently influenced pedicle dimensions in a statistically significant fashion, with male, taller, and heavier individuals having larger pedicles. However, the difference in pedicle dimensions among different age groups and between Caucasians and African Americans largely did not consistently reach statistical significance.

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TABLE 5. PH and PW of All the Sampled Specimens of This Study Compared With Those of Other Studies

Study Race	CS	Zindrick *et al*	Berry *et al*	Olsewski *et al*	Zindrick *et al*	Hou *et al*	Marchesi *et al*[27]	Mitra *et al*			
	American (N = 503)	American (N = 21–52)	American (N = 30)	American (N = 40–42)	Mean Values (N = 91–124)	Chinese (N = 40)	Swiss (N = 28–46)	Indian (N = 20)			
L1 PH	15.75	15.4	15.6	16.4	15.92	15.8	15.4	15.68			
L2	15.06	15	15.3	15.4	15.27	15.1	15	15.31			
L3	14.59	14.9	14.5	15.4	14.99	14.7	14.6	15.07			
L4	13.81	14.8	13.1	15.4	14.58	15	14.5	14.78			
L5	13.67	14	13.7	16.2	14.67	19.8	15.2	15.80			
Study Race	**CS**	**Zindrick *et al***	**Berry *et al***	**Olsewski *et al***	**Zindrick *et al***	**Hou *et al***	**Marchesi *et al***	**Mitra *et al***	**Chadha *et al***	**Krag *et al***	
	American (N = 503)	**American (N = 26–56)**	**American (N = 30)**	**American (N = 41–42)**	**Mean Values (N = 91–124)**	**Chinese (N = 40)**	**Swiss (N = 36–46)**	**Indian (N = 20)**	**Indian (N = 14–20)**	**American (N = 14–24)**	
L1 PW	8.52	8.7	7	8.2	7.96	7	7.1	6.94	6.69	7.01	
L2	8.66	8.9	7.5	8.4	8.28	7.4	7.9	7.72	7.26	8.67	
L3	10.46	10.3	9.2	10.2	9.99	9.2	9.6	8.86	8.43	9.3	
L4	13.19	12.9	10.4	13.2	12.32	10.5	13.4	11.37	10.81	11.03	
L5	18.33	18	10.7	20.1	16.98	12.9	18.8	15.85	13.47	15.15	

Direct caliper: CS, Berry et al, Olsewski et al, Hou et al, Marchesi et al, Mitra et al CT: Zindrick et al, Chadha et al, Krag et al.
PW indicates pedicle width; PH, pedicle height; CS, current study; CT, computed tomography.

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➢ Key Points

- ❑ PH decreased in size caudally down the lumbar spine, but PW increased in size. PW is the limiting factor in determining screw caliber selection due to its smaller size compared with PH.
- ❑ Sex, body height, and weight are 3 of the main factors that significantly contribute to the variation in lumbar pedicle dimensions of American population, with male, taller, and heavier individuals having larger pedicles.
- ❑ Age and race failed to consistently contribute to the differences in lumbar pedicle dimensions in a statistically significant manner.

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